

PhysicsMate: A Curriculum-Grounded Bengali Benchmark for Secondary Physics QA with Small-Model Adaptation

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Abstract—Bengali secondary education lacks curriculum-grounded benchmarks for STEM question answering, and general-purpose language models struggle with the precise terminology, unit conventions, and derivations that physics problems demand. We introduce PhysicsMate, a benchmark of 1,834 question-answer pairs built from the National Curriculum and Textbook Board (NCTB) Grade 9–10 physics syllabus and grounded in a multi-relational knowledge graph of 1,760 nodes and 2,600 edges across ten ontological types. We adapt Qwen3 models at 0.6B, 1.7B, and 4B parameters using Low-Rank Adaptation under a single fixed recipe and find that closed-book accuracy improves at every scale, with gains growing substantially at higher capacity (+5.5, +15.0, and +23.3 percentage points). A node-type analysis shows that structured curricular knowledge, such as physical quantities and named laws, benefits most from adaptation, while loosely specified entity-level knowledge benefits least. The adapted 4B model quantizes to a compact offline binary suitable for local inference in resource-constrained settings, offering a practical path toward curriculum-aligned physics assistance where connectivity and hardware are limited. The dataset, knowledge graph, and code will be made publicly available on Hugging Face upon acceptance.

Index Terms—Natural Language Processing, Low-Resource Languages, Knowledge Graphs, Question Answering, Small Language Models, Parameter-Efficient Fine-Tuning, STEM Education, Bengali.

I. INTRODUCTION

Bengali, spoken natively by over 300 million people, remains underrepresented in educational NLP. Existing Bengali benchmarks target sentiment analysis, news classification, and commonsense reasoning [1]; curriculum-aligned STEM resources for secondary education are scarce. BEnQA [2] and BnMMLU [3] broaden topical coverage but do not organize questions around a national curriculum, and P6Jiggasha [4] addresses physics one level above the syllabus considered here. As a result, a persistent gap remains between what current language models can produce and what classroom-aligned pedagogy actually requires.

Secondary physics in Bangladesh follows the NCTB Grade 9–10 syllabus, which spans 13 chapters covering mechanics, electromagnetism, optics, and modern physics. Answering correctly in this domain is not primarily a language-generation problem: it requires applying specific physical laws, stating definitions in SI units, performing dimensional analysis, and deriving numerical results in standard Bengali. General-purpose multilingual models frequently hallucinate dimensional formulas, produce code-switched text, and fall into repetitive

patterns on deterministic calculation questions, reflecting a gap that is ontological rather than purely linguistic: it demands structured internalization of physical relationships that base models do not encode by default.

No existing resource ties Bengali secondary-physics questions to curriculum structure in a way that supports controlled, category-level evaluation. As illustrated in Fig. 1, PhysicsMate addresses this gap with an end-to-end framework spanning curriculum ingestion, multi-relational knowledge formalization, QA benchmark construction, and parameter-efficient small-model adaptation:

- 1) **A curriculum-grounded benchmark and knowledge graph.** We formalize the 13-chapter NCTB syllabus as a multi-relational knowledge graph (1,760 nodes across 10 ontological types, 2,600 directed edges) and use it to construct 1,834 QA pairs in formal Bengali, partitioned into train, validation, and held-out test sets.
- 2) **A capacity-dependent adaptation study.** Under a single LoRA recipe, we adapt Qwen3 at three scales (0.6B, 1.7B, 4B). Accuracy gains grow monotonically with capacity (+5.5, +15.0, and +23.3 percentage points), and the adapted 4B model surpasses frontier cloud models on curriculum-aligned lexical and semantic metrics despite lower raw accuracy.
- 3) **An ontological diagnostic analysis.** Breaking results down by knowledge-node type shows that structured facts such as quantities, laws, and definitions adapt most readily, while entity-level knowledge adapts least, giving an empirical basis for targeted future improvement.

II. RELATED WORK

A. Bengali Educational and Science QA

BanglaBERT [1] established a pretrained encoder for Bengali. BEnQA [2] introduced parallel Bengali–English science questions and documented a substantial accuracy gap for GPT-4 relative to English. BnMMLU [3] spans 41 domains with over 134,000 multiple-choice questions but provides no ontological structure and does not target secondary physics specifically. P6Jiggasha [4] benchmarks Bangla physics at the Higher Secondary Certificate level using proprietary models, one level above Grade 9–10, without linking questions to curriculum structure. PhysicsMate targets Grade 9–10 physics explicitly and grounds every question in a curriculum-defined ontology.

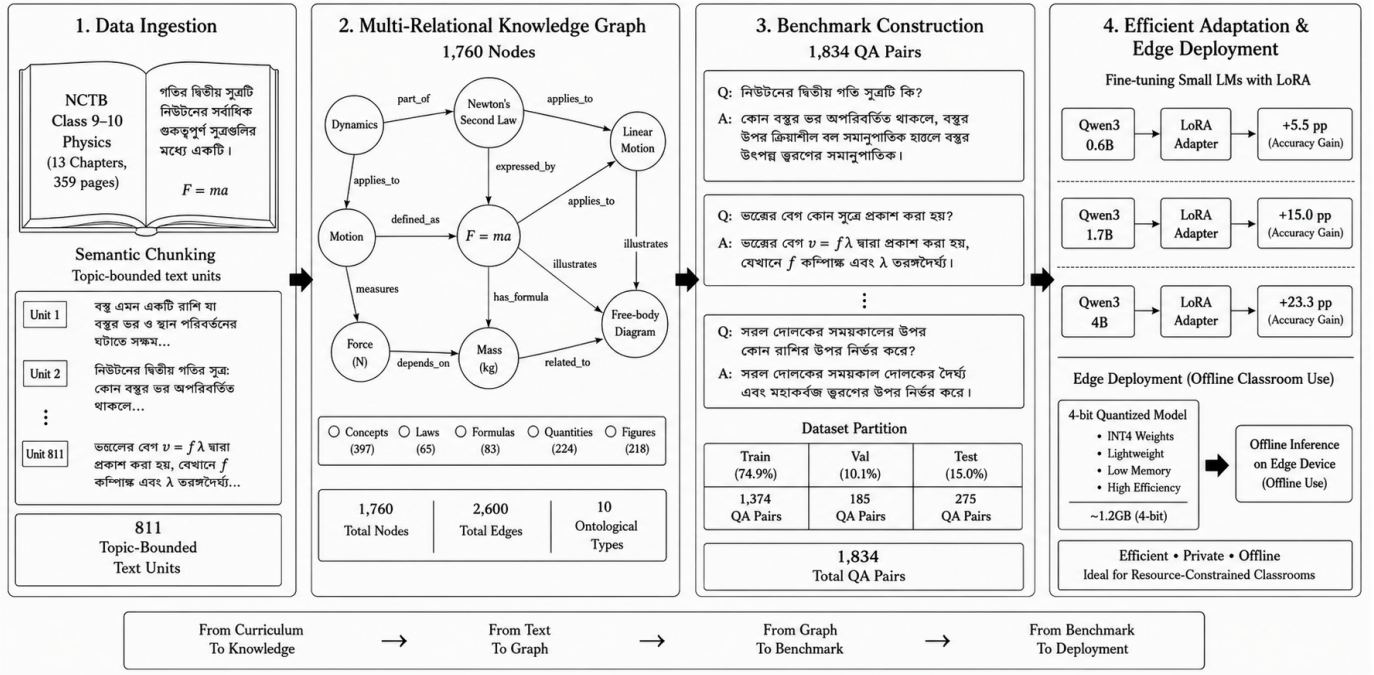


Fig. 1. The end-to-end PHYSICSMATE workflow. (1) **Data Ingestion**: page-level OCR of the NCTB Grade 9–10 physics textbook, Markdown normalization, and three-page contextual knowledge extraction yielding 811 semantic units; (2) **Multi-Relational Knowledge Graph**: a 10-type domain ontology with 1,760 nodes and 2,600 directed edges; (3) **Benchmark Construction**: 1,834 curriculum-grounded Bengali QA pairs stratified into Train/Val/Test; and (4) **Efficient Adaptation & Edge Deployment**: LoRA fine-tuning of Qwen3 models at 0.6B, 1.7B, and 4B parameters with 4-bit offline quantization.

B. Efficient Adaptation and Structured QA

The Qwen3 family [5] provides dense multilingual models from 0.6B to 32B parameters, and Paramanu [6] shows that compact Bengali-specific models under 400M parameters remain competitive on general tasks. LoRA [7] is the standard parameter-efficient adaptation method; Biderman et al. [8] report that it generalizes better outside the training distribution than full fine-tuning. On the structured-QA side, KG-for-PhysicsQA [9] generates knowledge graphs from physics questions for query decomposition, and Christou et al. [10] propose an OWL ontology for curriculum materials. These systems build graphs from text; PhysicsMate instead uses a curriculum-defined ontology to construct and partition the benchmark itself, enabling category-level evaluation.

III. THE PHYSICSMATE BENCHMARK

A. Curriculum Corpus and Semantic Chunking

PhysicsMate is derived from the official NCTB Grade 9–10 secondary physics textbook, covering 13 chapters and 359 pages (Fig. 1, Stage 1). We began with page-level optical character recognition (OCR) using Gemini 3.1 Flash Lite, converting each page into a normalized Markdown representation. This step preserves textual structure, mathematical expressions, and headings while removing the formatting noise that arises from direct PDF extraction.

The resulting Markdown pages were then processed using the same model in overlapping three-page context windows with a fixed domain-specific persona and a structured extraction prompt. For each window, the model identified self-contained

curricular knowledge units and assigned each to a predefined ontology schema covering concepts, physical quantities, figures, examples, entities, definitions, formulas, laws, and higher-level curricular topics. Providing three consecutive pages jointly as local context allowed knowledge units that span page boundaries to be captured as a single coherent record. Each unit retained its source-page provenance and topic association, making every knowledge record traceable back to the originating textbook material.

After extraction, the records were normalized and deduplicated, and topic boundaries were used to organize the corpus into 811 semantic units. Unlike fixed-length segmentation, each unit is designed to preserve one coherent curricular concept along with its associated mathematical notation, SI-unit conventions, worked examples, and figure references. These structured records served as the direct input to both the knowledge graph construction and the subsequent benchmark generation.

B. Knowledge Graph Architecture

The extracted curricular units were converted into a multi-relational knowledge graph using a fixed ontology designed for secondary physics (Fig. 1, Stage 2). The ontology captures complementary forms of curricular knowledge, including concepts, questions, physical quantities, figures, examples, entities, definitions, formulas, laws, and higher-level curricular topics. Each knowledge record was linked to its originating semantic unit, enabling source traceability from graph nodes back to the textbook.

TABLE I

KNOWLEDGE GRAPH STATISTICS. NODE TYPES CORRESPOND TO FOUR COGNITIVE LEVELS: DECLARATIVE RECALL (DEFINITIONS, LAWS), PROCEDURAL APPLICATION (FORMULAS, EXAMPLES), VISUAL INTERPRETATION (FIGURES), AND CURRICULAR SCAFFOLDING (HSC TOPICS).

Node Type	Count	Top Edge Type	Count
Concept	397	part_of	585
Question	252	applies_to	531
Quantity	224	illustrates	361
Figure	218	example_of	268
Example	188	depends_on	215
Entity	176	derived_from	118
Definition	136	+ 24 further relation types	
Formula	83		
Law	65		
HSC Topic	21		
Total nodes	1,760	Total edges	2,600

TABLE II

FIELDS STORED FOR EACH EXTRACTED KNOWLEDGE RECORD. EVERY RECORD IS ASSIGNED ONE ONTOLOGY TYPE AND RETAINS FULL PROVENANCE BACK TO THE SOURCE TEXTBOOK PAGES.

Field	Description
node_type	Ontology category (concept, law, formula, etc.).
title	Canonical name of the curricular item.
content	Self-contained Bengali description of the knowledge unit.
formula	Mathematical expression, when the item includes one.
unit	SI or textbook-specified physical unit, where applicable.
source_pages	Textbook page identifiers used as evidence.
topic	Chapter and topic within the NCTB curriculum.
relations	Typed links to related nodes (concepts, laws, figures, etc.).
provenance	Reference to the semantic unit of origin.

Relations were assigned according to the semantic role of each extracted record. The principal relation types encode curricular hierarchy (*part_of*), applicability of laws and formulas to physical situations (*applies_to*), visual grounding between figures and concepts (*illustrates*), example relations (*example_of*), dependency relations (*depends_on*), and provenance relations such as *derived_from*. This representation provides a structured curricular index over the source material and supports category-level analysis of the benchmark.

The resulting graph contains 1,760 nodes across 10 ontological types and 2,600 directed edges spanning 30 relation types. Table I summarizes the node and relation statistics.

To make the extraction procedure reproducible, Table II lists the fields that each knowledge record carries. Every record is typed according to the ontology, retains a canonical title and a self-contained Bengali description of the curricular item, and links back to the source pages from which it was drawn. Mathematical expressions, physical units, and typed relations to other nodes are stored alongside each record, giving the graph both factual content and explicit relational structure.

C. QA Construction and Partitioning

The benchmark items were derived from the structured knowledge records and their associated source contexts (Fig. 1, Stage 3). Each question–answer pair is anchored to a specific knowledge node and originating semantic unit, which provides explicit curricular provenance: every item can be traced

back to a particular textbook passage and its ontology-typed representation. The questions span all 13 chapters of the NCTB Grade 9–10 physics syllabus and are expressed in standard Bengali.

The final benchmark contains 1,834 question–answer pairs. Chapter 1 (Physical Quantities and Measurement) contributes the most items (302), while Chapter 12 (Magnetic Effects of Current) contributes the fewest (82). The corpus is partitioned into training ($n = 1,374$; 74.9%), validation ($n = 185$; 10.1%), and held-out test ($n = 275$; 15.0%) sets, stratified by chapter. Partition membership was verified programmatically so that no question identifier appears in more than one split. The average question length is 50.3 characters and the average answer length is 117.2 characters.

D. Data Provenance and Quality Control

The central design choice of PhysicsMate is that the NCTB textbook itself serves as the authoritative source for all content. Rather than generating questions from general knowledge, the extraction pipeline derives each benchmark item from a specific textbook passage, which is retained as metadata throughout the pipeline. This means every answer is expected to use the terminology, formulas, and unit conventions of the source curriculum rather than paraphrases drawn from other texts.

The fixed schema and structured prompt used during knowledge extraction reduce variability in how equivalent curricular facts are represented: the same physical law will consistently appear under its NCTB-designated name and formula rather than in a reformulated variant. Partition assignment was performed after the full benchmark was constructed, and each question identifier was checked against all three splits to confirm the absence of overlap. These properties together allow fine-tuning and evaluation to remain clearly separated and reproducible.

IV. EXPERIMENTAL SETUP

A. Models and Adaptation

We evaluate three Qwen3 scales: 0.6B, 1.7B, and 4B parameters. Each model is adapted using LoRA [7], applied to all seven linear projection layers per transformer block (query, key, value, output, gate, up, down), with rank $r = 16$, scaling factor $\alpha = 32$, and dropout $p = 0.05$, held fixed across all three scales. Models are trained for 3 epochs with 8-bit AdamW at a learning rate of 2×10^{-4} under cosine decay, with base weights loaded in 4-bit NF4 quantization; only the LoRA adapters are updated, approximately 0.8–1.7% of total parameters depending on scale. Since the adaptation recipe is identical across scales, model capacity is the primary variable distinguishing the three conditions.

B. Evaluation

All models are evaluated in closed-book mode on the 275 held-out test questions with greedy decoding, capped at 160 new tokens. We report three metrics: Token F1, a lexical overlap measure using a Bengali Unicode regex tokenizer; multilingual BERTScore [11] for paraphrase-tolerant semantic similarity;

TABLE III
RESULTS ON THE HELD-OUT TEST SET ($n = 275$). ACCURACY IS LLM-JUDGE (GEMINI-2.5-FLASH-LITE). CLOUD BASELINES ARE ZERO-SHOT; EDGE MODELS ARE EVALUATED WITH AND WITHOUT LoRA ADAPTATION.

Model	Params	Acc. (%)		Token F1		BERTScore	
		Raw	FT	Raw	FT	Raw	FT
<i>Cloud baselines (zero-shot)</i>							
GPT-4o-mini	>8B	69.5		0.239		0.728	
Gemini-2.5-FL	>10B	82.9		0.267		0.733	
<i>Edge models (local, offline)</i>							
Qwen3-0.6B	0.6B	0.0	5.5	0.216	0.283	0.696	0.753
Qwen3-1.7B	1.7B	10.5	25.5	0.227	0.299	0.706	0.766
Qwen3-4B	4.0B	27.6	50.9	0.252	0.366	0.722	0.783

and LLM-as-a-Judge accuracy [12], where Gemini-2.5-Flash-Lite at temperature 0 marks each prediction correct only if it conveys the gold reference’s key fact. Cloud baselines (GPT-4o-mini, Gemini-2.5-Flash-Lite) are assessed zero-shot with the same Bengali prompt and test questions used for the local models.

V. RESULTS

A. Main Benchmark

Table III reports results across all conditions. Fine-tuning improves every metric at every scale.

The base Qwen3-0.6B produces no correct answers on the test set; after adaptation it reaches 5.5% accuracy, confirming that even a sub-billion-parameter model acquires some domain grounding. The 1.7B model more than doubles its accuracy, from 10.5% to 25.5%, and the 4B model achieves the strongest edge performance at 50.9%, a gain of 23.3 percentage points over its own raw baseline. Relative Token F1 improvements follow the same pattern, increasing from 31.0% at 0.6B to 31.7% at 1.7B and 45.2% at 4B.

Cloud baselines achieve higher raw accuracy, 69.5% and 82.9% respectively, but their Token F1 and BERTScore fall below the adapted 4B model’s. This gap arises because the cloud models tend to produce longer, more explanatory responses than the concise, NCTB-standard phrasing that the reference answers use, which the lexical and semantic metrics penalize even when the underlying reasoning is sound.

B. Capacity-Dependent Adaptation

Figure 2 shows that adaptation gains grow with model capacity. The accuracy improvement trajectory is:

$$\Delta \text{Acc} : +5.5 \text{ pp} \rightarrow +15.0 \text{ pp} \rightarrow +23.3 \text{ pp} \quad (1)$$

The step from 1.7B to 4B yields a 25.4 percentage point higher fine-tuned accuracy (reaching 50.9% from 25.5%), indicating that larger base models make more effective use of the same LoRA capacity to internalize domain-specific structure. With three data points this constitutes an observed trend rather than a formal scaling law, but the monotonic pattern holds consistently across accuracy, Token F1, and BERTScore alike.

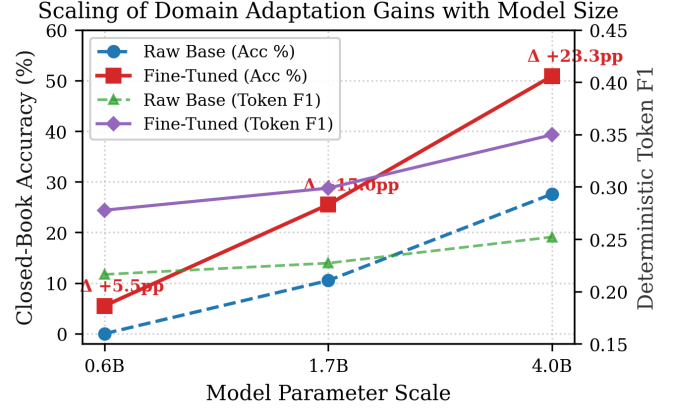


Fig. 2. Capacity-dependent adaptation gains. Fine-tuning produces monotonically increasing accuracy improvements (+5.5, +15.0, and +23.3 percentage points) and consistent gains in Token F1 and BERTScore as model size grows from 0.6B to 4B parameters.

TABLE IV
QWEN3-4B TOKEN F1 BY KNOWLEDGE-NODE TYPE ($n = 275$ TEST QUESTIONS).
STRUCTURED KNOWLEDGE TYPES BENEFIT MOST FROM ADAPTATION.

Node Type	n	Raw F1	FT F1	Rel. Gain
Quantity	29	0.262	0.508	93.9%
Law	8	0.286	0.448	56.6%
Definition	28	0.228	0.356	56.1%
Example	35	0.244	0.358	46.7%
Concept	65	0.250	0.342	36.8%
Figure	67	0.263	0.320	21.7%
Formula	8	0.280	0.332	18.6%
Entity	35	0.241	0.262	8.7%

C. Node-Type Diagnostics

Table IV breaks down Qwen3-4B Token F1 by knowledge-node type on the test set. Quantity nodes show the largest relative gain (93.9%), followed by scientific laws (56.6%) and definitions (56.1%). Entity nodes show the smallest gain (8.7%).

The ordering is interpretable: quantities, laws, and definitions are concrete, well-structured facts with canonical Bengali phrasings in the NCTB textbook, while entity mentions are loosely specified and often admit multiple valid surface forms. Low-rank adaptation captures the former more readily than the latter, matching the capacity-dependent pattern observed in Section V-B.

D. Qualitative Analysis

Table V presents two representative cases illustrating failure modes that adaptation corrects.

In the optics case, the base model fabricates a spatial rule not present in the curriculum, while the adapted model produces the correct three-adjective answer. In the circuits case, the base model recites Ohm’s Law but never completes the algebraic step to the power formula; the adapted model states $P = I^2 R$ directly. Both patterns match the node-type results above: adaptation improves most where a single canonical answer exists.

TABLE V
REPRESENTATIVE BASE VS. FINE-TUNED QWEN3-4B OUTPUTS. FINE-TUNING CORRECTS FACTUAL HALLUCINATION AND INCOMPLETE DERIVATION.

Category	Gold Reference	Base Qwen3-4B	Fine-Tuned Qwen3-4B
Optics (ch09_q084) <i>Image properties for an object between a convex lens and its focus?</i>	প্রতিবিম্ব অবাস্তব, সোজা ও বস্তুর তুলনায় বড়ো হয়। [Virtual, erect, magnified.]	<i>Fabricated rule:</i> প্রতিবিম্ব বামে দাঁড়ায় (বস্তুর বাম দিকে)... [Invents a spatial rule absent from the curriculum.]	প্রতিবিম্ব অবাস্তব, সোজা ও বস্তুর তুলনায় বড় হয়। [Matches gold reference.]
Circuits (ch11_q031) <i>Express power in terms of current and resistance.</i>	$P = I^2 R$ [Power dissipated as heat in the resistor.]	<i>Incomplete:</i> $V = I \times R$, যেখানে $V = P = I^2 R$ বা $P = V^2 / R$ দ্বারা ক্ষমতাকে প্রকাশ করা যায়। ভোল্ট, $I =$ প্রবাহ... [Stops at Ohm's law; never derives the power expression.]	

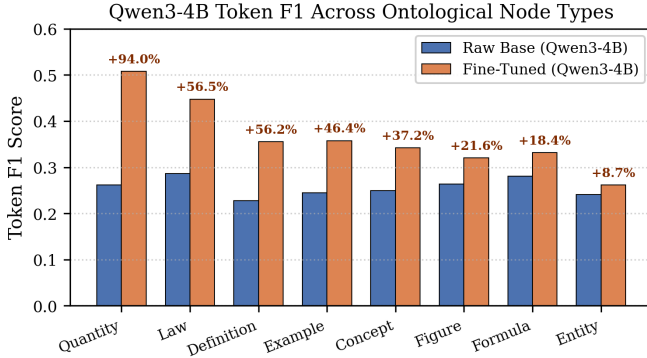


Fig. 3. Token F1 by knowledge-node type for base vs. fine-tuned Qwen3-4B. The largest gains appear on structured physical quantities (93.9%) and scientific laws (56.6%); entity-level knowledge shows the smallest improvement (8.7%).

VI. DISCUSSION, LIMITATIONS, AND CONCLUSION

Three observations emerge from these experiments. First, parameter-efficient adaptation materially improves Bengali secondary-physics QA at every tested model scale, even for a base model that produces no correct answers before adaptation. Second, the magnitude of improvement depends strongly on capacity: the same LoRA configuration yields a 23.3 percentage point gain at 4B but only 5.5 at 0.6B, suggesting that larger models provide more usable representational space for domain-specific structure. Third, the gains are uneven across curricular knowledge types, concentrating in structured categories such as quantities, laws, and definitions where canonical phrasings exist, and diminishing for loosely specified entity knowledge.

The adapted 4B model remains well below frontier-model accuracy, 50.9% versus 82.9%, and we do not claim that PhysicsMate solves Bengali physics QA. What the results establish is that curriculum-grounded adaptation is effective, and that the benchmark provides the diagnostic granularity needed to identify where further improvement is required. The 4B model quantizes to a compact offline binary suitable for local inference, a property that matters directly for deployment in settings with limited connectivity.

Several limitations apply. The evaluation is text-only; the 218 figures linked in the knowledge graph are not used as visual inputs. Accuracy relies on an LLM judge rather than human annotation. The held-out split is disjoint by question ID, though we do not claim a broader contamination analysis beyond that check. The benchmark covers a single national

curriculum, NCTB Grade 9–10 physics, and generalization to other syllabi or subjects has not been tested.

PhysicsMate provides a curriculum-grounded Bengali secondary-physics benchmark, a controlled adaptation study across three model scales, and an ontological diagnostic that identifies where parameter-efficient adaptation helps most. Together, these components offer an empirical foundation for developing and evaluating efficient Bengali educational language models.

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